

DEXTER FERNANDES

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SUMMARY

Computer Vision Engineer with 3+ years building and deploying **real-time** vision systems on **edge** hardware for safety-critical industrial sites. Owned production pipelines end to end on NVIDIA DeepStream and OpenVINO for oil and gas and construction safety clients, from data curation and training through **quantisation**, deployment and **on-site validation**. Led a four-engineer team on a construction safety detector, cutting false positives by 33% within latency and power budgets. MSc Robotics (Merit) with **LiDAR SLAM**, point cloud and multi-sensor calibration work in C++17 and ROS2.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end **computer vision** lifecycle for **real-time** perception on BP drilling rigs, from data curation and training through deployment and post-deployment monitoring, improving F1 by 15% under 24/7 operation.
- Developed and maintained NVIDIA **DeepStream** 6.x and **FFmpeg** video pipelines covering camera ingestion, inference and field diagnostics in desert conditions.
- Hardened deployed services with **Docker** microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Built a watchdog with a REST API exposing service health and restart control, and added **observability** through structured logs, metrics, timeouts and retries that cut **root-cause analysis** time on live deployments.
- Led **on-site validation** on BP rig deployments, diagnosing model failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.
- Adapted YOLOv5 detectors in **PyTorch**, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and Weights & Biases tracking across runs.
- Applied image preprocessing and artefact detection with **OpenCV** and scikit-image to catch mud and sand contamination on camera feeds, reducing false positives by 13%.
- Built ingestion, labelling and validation pipelines across image, video and sensor data with Airflow, MLflow, CVAT and DVC **dataset versioning**, processing 3.6 million records and cutting rollout from 72h to under 24h.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage **object detection** model feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Built asynchronous **C++17** perception pipelines on **OpenVINO** across Intel Movidius VPUs and Core i7 industrial edge boxes, raising throughput from 5 to 15 FPS under real-time constraints.
- Delivered a real-time multi-camera **video analytics** pipeline on a moving construction vehicle, synchronising streams and adding ByteTrack multi-object **tracking** to hold worker and vehicle identities, cutting duplicate incident alerts.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in **TensorFlow Keras** and OpenVINO, cutting model size by 50% while holding production accuracy.
- Developed YOLOv5 visual quality inspection for John Deere engine paintjobs on a conveyor line, detecting paint defects, contamination and colour inconsistency.
- Tuned camera and ISP settings to keep models reliable on RGB and IR streams across 24/7 operation, cutting false positives on IR and reducing manual recalibration.
- Used confusion-matrix failure clustering and TIDE error analysis on COCO-format production data to diagnose recurring detector failures under **domain shift**, prioritising dataset fixes across camera feeds.
- Set experimentation standards, reviewed model performance and **mentored** junior engineers, guiding optimisation strategy for production deployments.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Developed an end-to-end DeepStream 9.0 and **GStreamer RTSP** pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a dynamic-batch **TensorRT** FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream, with a C++ and CUDA decode plugin moving post-processing into the engine.
- Designed a geometry-first cross-camera identity layer over per-camera trackers, fitting uncertainty-aware ground-plane **homographies** behind a held-out quality gate and lifting pooled IDF1 from 0.198 to 0.245 on WildTrack.
- Added structured JSON-line logging and a per-sensor health monitor flagging stalled streams, and gated model promotion on match-rate and mean-IoU against a signed SHA-256 manifest inside a GitHub Actions **CI/CD** workflow.

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in **ROS2** Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D **LiDAR** configurations, comparing mapping and localisation behaviour to support system selection.
- Implemented **point cloud** filtering, segmentation and registration with **PCL** and Open3D, and added an EKF localisation layer with robot_localization fusing wheel odometry and IMU.

- Developed ROS2 nodes, launch files and parameterised configs in C++ and **Python**, using tf2 and message synchronisation to debug transform timing, and containerised the stack with **Docker** on Linux.
- Modern Edge CV Benchmarking** github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking 03/2026
- Benchmarked modern **object detection** models including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
 - Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess **segmentation** and depth models under the same constraints, exporting to **ONNX** and testing OpenVINO deployment paths.

EDUCATION

- University of Bristol** — MSc Robotics, Merit 09/2023 – 09/2024 · BRISTOL, ENGLAND
- Dissertation: Radiological SLAM with LiDAR Odometry
- Built a real-time SLAM system in C++17 and **ROS2** (rclcpp) on a Clearpath Husky, fusing Velodyne VLP-16 **LiDAR**, RealSense D435i depth and radiological sensing for inspection mapping.
 - Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a **PCL point cloud** pipeline, validating sensor placement for coverage.
 - Ran the full SLAM and perception stack on an NVIDIA **Jetson AGX Xavier** under JetPack, working within the platform's compute and memory limits for on-robot operation.
 - Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol, configuring Ceres, g2o and GTSAM as solver backends and tuning ICP parameters.

- Visvesvaraya National Institute of Technology** — BTech Electrical and Electronics Engineering 07/2016 – 05/2020 · NAGPUR, INDIA
- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash, SQL
Computer Vision	Object detection, image classification, semantic and instance segmentation, object tracking, anomaly detection, defect detection, optical flow, video analytics, data augmentation, CNNs, YOLO (v5/v8/v10/v26), RT-DETR, EfficientDet, ByteTrack
Deep Learning	PyTorch, TensorFlow, Keras, OpenCV, scikit-image, torchvision, NumPy, HuggingFace, EfficientViT, MobileSAM, Depth Anything v2
Video Pipelines	NVIDIA DeepStream 6.x/9.0, GStreamer, FFmpeg, RTSP, nvinfer, nvstreammux, nvtracker, NvDCF, NvSORT, multi-stream video analytics, multi-camera synchronisation, multi-target multi-camera (MTMC) tracking
Camera and 3D	Camera geometry, camera calibration, homography calibration, ground-plane projection, LiDAR extrinsics calibration, ISP and camera tuning (RGB and IR), LiDAR, point clouds, PCL (Point Cloud Library), Open3D, depth sensing
Edge Inference	TensorRT (custom plugins, dynamic batching), OpenVINO, ONNX, quantisation-aware training, post-training quantisation, FP16/INT8, structured pruning, layer fusion, latency and throughput profiling, Intel VTune, NVIDIA Jetson AGX Xavier, Intel Movidius VPU, industrial edge compute
Robotics	ROS2 (Robot Operating System 2), rclcpp, rclpy, tf2, rosbag2, Gazebo, Nav2, RTAB-Map, LIO-SAM, Cartographer, GLIM, LiDAR SLAM, ICP (Iterative Closest Point), robot_localization, EKF, Eigen
MLOps and Data	MLflow, DVC (Data Version Control), Apache Airflow, Apache Kafka, Optuna, Weights & Biases, CVAT, automated labelling, dataset versioning, synthetic data generation, benchmarking harnesses, ablation studies, TIDE, COCO annotation format, drift monitoring, model-promotion gates
Production	Linux, Docker, CI/CD, GitHub Actions, Git, CMake, pytest, unittest, gtest, REST API design, containerised microservices, multiprocessing, Microsoft Azure, PostgreSQL, structured logging, health monitoring, observability, Agile and Scrum, technical leadership, mentoring

ADDITIONAL EXPERIENCE

- GrowthStage** — Technical Assessor, Computer Vision and Software Engineering 05/2026 – PRESENT · REMOTE
- Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.
- Taco Bell** — Service Champion / Team Trainer 09/2024 – PRESENT · BRISTOL, ENGLAND
- Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.

CERTIFICATIONS AND AWARDS

- Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)